TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Detailed Study of Relevant Mechanisms and Machinery of Advanced Weaving

A guided dissection of the shuttleless loom as a system: shedding, weft insertion, beat-up, take-up and let-off, and the auxiliary mechanisms — with each motion's timing recorded against the loom's crank circle.

FM-4102 · ADVANCED WEAVING TECHNOLOGY-II SESSIONAL · EXPERIMENT 1 · FABRIC MANUFACTURING SPECIALIZATION

1. Aim and theory

Aim: to identify, sketch and explain every principal mechanism of the laboratory's advanced (shuttleless) loom, and to record its timing and production data.

A loom is a timed machine: all its motions are phases of one revolution of the main shaft (360° per pick). The cycle's fixed point is **beat-up at 0°** (reed at the cloth fell). Against that reference: the shed must be fully open before the weft carrier enters, the carrier must clear the shed before it closes, and take-up and let-off creep continuously or intermittently to keep the fell position constant. Advanced looms replace the shuttle with **projectile, rapier or fluid-jet insertion** and progressively replace mechanical drives with individual motors and electronic control — but the timing logic is unchanged, which is why this study records *angles*, not just parts.

2. Apparatus

- Laboratory rapier/air-jet loom (stopped, with guards removable for study), crank-angle timing disc on the main shaft, stroboscope or tachometer, steel rule and feeler gauges, weft package and pirn for comparison, loom manual and setting charts, sketchbook.

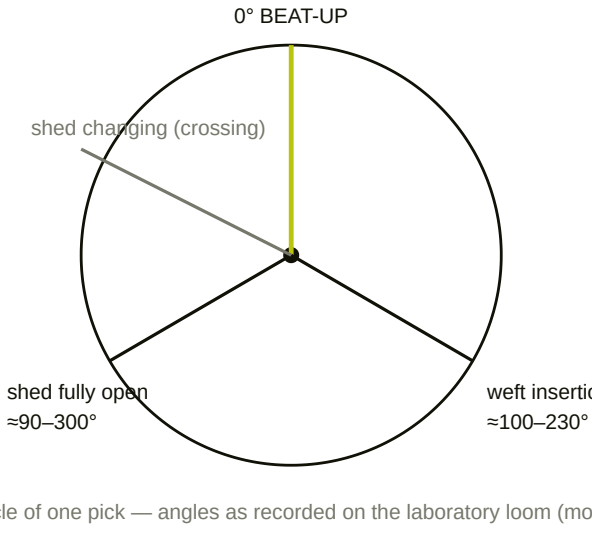


Figure 1 — Loom timing circle: every event of the pick cycle is an angle of the main shaft measured from beat-up. The study method is to find each event's angle on the timing disc and compare it with the maker's chart.

3. Procedure — mechanism by mechanism

1. **Shedding system:** identify the type fitted (cam, dobby or jacquard unit). Record: number of shafts, shed geometry (shed height at the reed, measured with rule/gauge), shed timing (angle at which the shed is level/crossing, read from the timing disc), and the drive path from main shaft to heald frames. Sketch the shed geometry at full opening.
2. **Weft insertion system:** identify the carrier (projectile / rapier type / jet nozzles). Trace the weft path from package through accumulator/feeder, tensioner, selector, carrier/nozzle, across the shed to the receiving side and cutter. Record the insertion window from the timing disc. For a jet loom, note main and relay nozzle arrangement and air pressure settings; for rapier, note giver/receiver handover at shed centre.
3. **Beat-up:** identify the sley drive (crank or cam), reed number and reed width; record reed count from the reed itself (dents per 10 cm counted directly) and the number of ends per dent from the warp in the reed.
4. **Take-up and let-off:** identify take-up type (mechanical gear train with change wheel, or electronic/servo) and how picks/cm is set; identify let-off type (negative/positive, warp tension sensing) and trace how fell position is kept constant.
5. **Auxiliaries:** weft stop motion, warp stop motion (drop wires), weft selector/mixer, selvedge devices (leno or tuck-in), temple type, cloth wind-up, and the loom's monitoring (pick counter, efficiency/stop display). For each, state what fault it prevents or detects.
6. **Running record:** with the loom running, measure speed with the tachometer, record the set picks/cm, and observe insertion and beat-up under the stroboscope.

4. Worked calculation (model data)

PRODUCTION AND SETTING ACCOUNT OF THE STUDY LOOM

Loom speed measured: **600 picks/min** (air-jet, laboratory setting). Set density: 20 picks/cm. Theoretical production = 600 × 60 ÷ (20 × 100) = 36,000 ÷ 2,000 = **18.0 m/h**; at the recorded 85% running efficiency ≈ **15.3 m/h**.

Reed check: ends counted in reed = 3600 ends across 150 cm at 2 ends per dent → dents = 3600 ÷ 2 = 1800 → reed number = 1800 ÷ 150 = **12 dents/cm** — to be confirmed by counting dents over 10 cm of the reed itself (Experiment discipline: the counted reed, not the data sheet, is the record).

5. Observation table (as recorded)

ITEM	RECORD
Loom type / insertion	Air-jet, main + relay nozzles; weft accumulator with electronic feeder
Shedding	Electronic dobby, 16 shafts fitted; shed crossing ≈ 300° (level), full open through insertion window
Insertion window	≈ 100°–230° of main shaft (timing disc)
Reed	12 dents/cm counted; 2 ends/dent; reed width 150 cm
Take-up / let-off	Electronic take-up (picks/cm set at panel, 20/cm); positive let-off with warp tension sensor
Stop motions	Piezo weft sensor; electric warp stop with drop wires; leno selvedge devices both sides
Speed / efficiency	600 picks/min; 85% over the observation period

Model data shown — in the laboratory each value is replaced by the student's own reading; the conclusions below are the form the discussion should take.

6. Discussion points

- Why the shed must be fully open *before* 100° and cannot close before ≈230°: the carrier's flight occupies the window; early closing traps or breaks the pick.
- Electronic take-up/let-off removes the change wheel but not the physics: picks/cm is still cloth length divided into picks — faults in tension sensing reappear as pick-spacing bars (see Experiment 3).

- What each auxiliary costs when it fails: a blind weft sensor passes broken picks into the cloth; a slack leno device gives a wavy selvedge that no finishing can cure.

7. Precautions

- All internal study is done with the loom **stopped and isolated**; guards removed only under supervision, replaced before running.
- Never reach into the shed or reed path with the machine live — turn by hand (inching) only, in the maker's direction.
- Record angles from the disc with the reference (0° = beat-up) stated; an angle without its reference is not data.

8. Practice (viva preparation)

1. Why is beat-up, not shed opening, chosen as 0° of the loom cycle?
2. A loom runs at 750 picks/min inserting 22 picks/cm at 88% efficiency. Find its production in metres per 8-hour shift.
3. State the function of the weft accumulator on a jet loom.
4. The reed count counted directly disagrees with the data sheet. Which do you record for the experiment, and why?
5. Name two faults that appear if let-off tension control hunts (oscillates).

Answer key

1. Beat-up is the one event fixed by the cloth itself — the reed driving the pick home at the fell — and it is mechanically unmistakable on any loom design; shedding and insertion timings differ by machine type and are therefore best expressed relative to it.
2. Production = 750 × 60 ÷ (22 × 100) = 45,000 ÷ 2,200 ≈ 20.45 m/h; per shift at 88%: 20.45 × 8 × 0.88 ≈ **144 m**.
3. It stores a measured length of weft under controlled low tension and releases it to the jet on demand, decoupling the violent, intermittent insertion from the package — without it, package unwinding tension spikes would break or slacken every pick.
4. The counted reed. The experiment's purpose is the machine as it is; the sheet may describe another reed, a past change, or an error — the count over a measured length is primary evidence.
5. Warp tension oscillation moves the cloth fell, giving **pick-spacing variation (weft bars/thin-and-thick places)** and, at the extreme, **warp breaks and slack ends** from tension peaks — both graded as fabric defects in Experiment 3.